

Changing demographics and young adult driver license decline in Melbourne, Australia (1994–2009)

Alexa Delbosc · Graham Currie

Published online: 17 August 2013
© Springer Science+Business Media New York 2013

Abstract In the last decade young people in North America, Australia and much of Europe are becoming less likely to hold a driver's license and, if they can drive, they are driving less. This is a remarkable trend which is not yet well understood. This paper is an empirical analysis exploring the relationship between young adult driver licensing and young adult demographics and living arrangements. In many developed countries, young adults are becoming increasingly less likely to be in full-time work, more likely to be in part-time work or studying, more likely to be living at home with parents and they are getting married and having children later in life. Against the background of these trends, a binary logistic regression model of travel survey data (1994–2009) for Melbourne, Australia is used to explore the association between these demographics and young adult license-holding. The model established that full-time employment and child-rearing are associated with higher young adult licensing rates whereas part-time work and studying were associated with lower licensing rates. However the impact of living at home with parents was not clear and requires further study. Together it is theorised that these changes in living arrangements may be restricting the disposable income of some young adults and reducing or postponing license take-up. The paper concludes with the implications of findings for policy and opportunities for future research.

Keywords Car licensing · Young adults · Demographic trends

A. Delbosc (✉) · G. Currie
Institute of Transport Studies, Department of Civil Engineering, Monash University, Building 60,
Clayton, VIC 3800, Australia
e-mail: alexa.delbosc@monash.edu

G. Currie
e-mail: graham.currie@monash.edu

Introduction

In recent decades, several industrialised nations have recognised a trend of stagnating or decreasing car travel demand (Zumkeller et al. 2004; Puentes and Tomer 2008; Le Vine et al. 2009; Newman and Kenworthy 2011; Kuhnimhof et al. 2012). At the same time a complementary trend is emerging: young people in North America, Australia and parts of Europe are less likely to initiate car-driving and, if they can drive, they are driving less (Raimond and Milthorpe 2010; Kuhnimhof et al. 2011; Sivak and Schoettle 2011; Sivak and Schoettle 2012).

Although there is a large body of research on understanding mode choice and car ownership more generally (e.g. Scheiner and Holz-Rau 2007), the downward trend in car travel amongst the young has only recently been recognised. Multiple causes have been explored, including household incomes, car availability, changing cultural attitudes, economic trends, the cost of motoring, graduated licensing and increased inner city living (Noble 2005; McDonald and Trowbridge 2009; Raimond and Milthorpe 2010; Davis et al. 2012; Licaj et al. 2012). However, very few studies have considered the influence of another cultural trend: the gradual postponement of the transition from studying and living at home to full-time employment and having children (Cobb-Clark 2008; Australian Bureau of Statistics 2012; Le Vine and Jones 2012).

This paper explores the relationship between a range of household living arrangements and the driver licensing behaviour of young adults. It seeks to explore these issues to better understand the downward trend in licensing occurring for young people. The focus of the paper is the population of Melbourne, the capital of the state of Victoria, Australia. Travel survey evidence is explored using an empirical analysis exploring household living arrangements and their links to driver licensing.

The paper takes the following structure: the following section describes the research to date on driver's licensing trends. It is followed by a description of the survey data and analysis methods. Then the results of comparative analyses and logistic regression modelling are presented and discussed.

Research context

Whether someone holds a driver's license or not has a significant impact on their mode choice and, consequentially, has flow-on impacts on road congestion and public transport use. In the past, research on driver licensing has focussed on the road safety implications of young drivers and variations in licensing schemes (e.g. Preusser et al. 1998). Yet after decades of increased take-up of driver licensing, in recent years researchers have begun to identify a downward trend in driver's licensing among young people. For example in the United States the proportion of 18-year-olds with a license increased from 60 % in 1963 to 80 % in 1983 (United States Census 2004; Federal Highway Administration 2011) but dropped back to 65 % in 2008 (Sivak and Schoettle 2011). Australia is showing a similar trend in recent years (see Fig. 1); in the state of Victoria, license-holding dropped over 1 % per year for people up to 30 (VicRoads 2012). Even in developed countries where licensing has not reduced, young adults are significantly reducing their automobile travel and increasing their use of alternative modes (Kuhnimhof et al. 2011; Kuhnimhof et al. 2012).

Emerging research is attempting to understand the factors influencing this trend, considering such influences as household incomes, economic trends, car availability, changing cultural attitudes, graduated licensing and urban form (Noble 2005; McDonald and Trowbridge 2009; Raimond and Milthorpe 2010; Davis et al. 2012; Delbosc and Currie

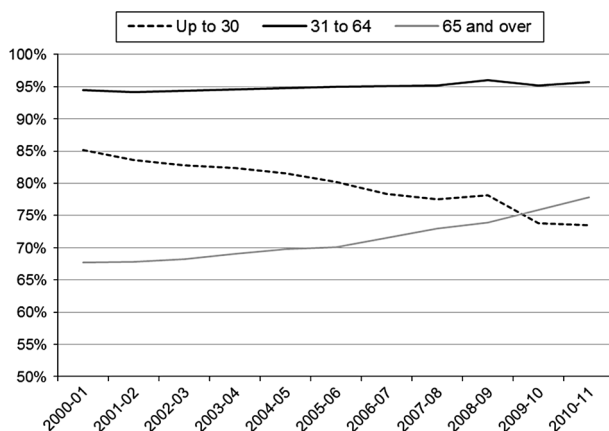


Fig. 1 Licensed drivers as a percentage of age group, 2000/2001–2010/2011, Victoria, Australia. *Note* data sourced by Australian financial year (July to June). *Source* licensing data supplied by VicRoads (2012) and compared to population (Australian Bureau of Statistics 2010)

2012; Licaj et al. 2012; Delbosc and Currie 2013b). This literature, however, is limited in two ways: it rarely combines explanatory variables with an exploration of wider societal trends and it has not yet isolated the influence of life stage on licensing.

Most of the emerging research literature into youth licensing influences does not discuss how those influences may or may not contribute to trends over time (e.g. McDonald and Trowbridge 2009; Licaj et al. 2012). Conversely, the studies that do examine the long-term trends in licensing are often speculative as to what might be causing the trend (Raimond and Milthorpe 2010; Kuhnimhof et al. 2011). Although the longitudinal nature of this study is limited, this paper presents an empirical analysis of travel survey data from 1994 to 2009 to both identify the degree to which licensing has changed over time and establish the likely causes of this change.

Four trends in living arrangements are the main focus of this research: a decrease in full-time youth employment, an increase in young adults enrolled in post-secondary education, a postponement in marriage and child-rearing and an increase in young adults living with parents. Out of all the possible social trends, these trends have been identified as among the most likely to directly impact youth licensing (Delbosc and Currie 2013b).

Changes in employment and education

In many developed countries, young people are becoming less likely to be in full-time employment, more likely to be in part-time employment, and more likely to be enrolled in post-secondary (i.e. post-high school) education. For example, in America the employment-population ratio for young adults (age 16–24) has decreased from a high of 67 % in 1990 to 49 % in 2010 (Bureau of Labor Statistics 2011). In Australia, the proportion of young adults in full-time work has dropped from almost 50 % in 1985 to 33 % in 2011 whereas in that time part-time work has steadily increased from 11 to 29 %.

This shift toward part-time employment is likely related to similar trend toward growing enrolments in post-secondary education. In Australia, the percent of people aged 20–24 attending some form of education increased from 25 % in 1991 to 41 % in 2011 (Australian Bureau of Statistics 2011). In America, enrolment in high school or university

increased from 31 % in 1990 to 45 % in 2011 (Taylor et al. 2012) and in the UK, full-time educational enrolment increased from 10 % in 1992 to almost 20 % in 2011 (Office for National Statistics 2011).

Taken together, these trends likely act to restrict the disposable income available to young adults. Low income, in turn, is associated with lower car licensing and ownership (Licaj et al. 2012).

Postponement of cohabitation and child-rearing

Young adults have increasingly been delaying or forgoing the transition into partnerships and child-rearing. In Australia, the proportion of people living with a partner has decreased steadily; for example the proportion of women aged 20–24 living with a partner dropped from 39 % in 1986 to 24 % in 2001 (Gray et al. 2008). So too the median age at first marriage has increased 3 years between 1990 and 2010 and the median age of first-time mothers has increased from 27.5 to 28.9 (Australian Bureau of Statistics 2012). Past research has established a clear link between having children and owning a car (Prevedouros and Schofer 1992; Ryan and Han 1999; Woldeamanuel et al. 2009). Similarly, starting family may be a motivator to get a driver's license and delaying this life stage may reduce the pressure on young adults to get a license.

Increase in living with parents

In addition to the previous trends, in the last few decades there has been trend in many countries toward young adults living at home with parents for longer (Cobb-Clark 2008). For example in Australia the proportion of people aged 20–34 living at home increased from 19 % in 1986 to 23 % in 2006 (Australian Bureau of Statistics 2009a). For those aged 20–24 living at home with parent, almost half said they did so for financial reasons; for those aged 25–29 this dropped to 20 % with another 20 % saying they remained for 'convenience or enjoyment' (Australian Bureau of Statistics 2009a). Living with parents may influence licensing rates although the direction of influence is unclear. Living at home is likely to provide financial support, access to experienced drivers and access to household vehicles; all of these influences are likely to increase the ability of young people to attain a driver's license. However if parents are providing transportation for their adult children this creates a disincentive for a young person to obtain their own license. Research from Sydney, Australia provides an initial suggestion that living at home may decrease licensing, as more trips by young people are being made as a vehicle passenger (Raimond and Milthorpe 2010). However an analysis of teenagers in the United States found no consistent relationship between parental living arrangements and licensing rates, leaving this an open question (Preusser et al. 1998).

Isolating the relationship between these life stages—transitioning from education to employment, moving away from home and having children—and young adult licensing is the primary contribution of this paper.

Methodology

A quantitative analysis of travel survey records was undertaken to explore the relationship between driver licensing trends for young people and living arrangement patterns. Analyses were conducted in two stages:

1. First, a series of descriptive analyses identified general trends in licensing and demographic changes between the four survey years.
2. Then a binary logistic regression was conducted to identify the degree of association between each predictor variable on whether or not a young person had a license. This method was selected as similar methods have been used in past studies of licensing (McDonald and Trowbridge 2009; Licaj et al. 2012).

Although in this model license-holding is conceptualised as the dependent variable, it should be recognised that causality cannot be unconditionally established. Rather, the analysis will establish associations between variables.

These analyses focus on young adults aged 18–30 in Melbourne, Australia. Eighteen is the age when young adults can legally drive independently and internationally, declines in licensing tend to be observed up until age 30 or even later (Sivak and Schoettle 2012; Kuhnimhof et al. 2012).

Travel survey data

The data for this study are drawn from household travel surveys conducted in Melbourne, Australia. From 1994 to 1999 this survey was called the Victorian Activity Travel Survey (VATS); after a hiatus of several years a similar survey called the Victorian Integrated Survey of Travel and Activity (VISTA) was instituted in 2007 and 2009. For VATS, over 5,000 private residential households across the greater Melbourne area were sampled each year; for VISTA this was over 10,000. VATS used a mail-out/mail-back self-complete questionnaire supplemented by phone calls in the case of questionable data and non-response interviews. VISTA hand-delivered and hand-collected the self-complete questionnaires, supplemented by reminder phone calls and letters. Otherwise the survey was designed to be as comparable as possible to VATS. Both surveys collected information about household and individual travel, demographics and living arrangements. Although these surveys included samples outside of Melbourne, these regional samples varied year on year so only the Melbourne data was analysed.

People aged 18–30 in the survey years of 1994, 1999, 2007 and 2009 were the focus of this analysis; this resulted in an unweighted sample size of 14,818 individuals. The initial descriptive analysis was sub-divided the survey sample into two age groups, 18–23 and 24–30, as living arrangements tend to change significantly during young adulthood.¹

Variables considered in analysis

Key variables examined include:

- Household vehicles—People were classified as living in either a household without any cars, with less than one car per adult, or with one or more cars per adult. A lack of household cars is likely to make it difficult to practice supervised driving, yet this variable is also likely to be highly endogenous with licensing. For this reason two logistic regressions were run, one including household vehicles and one without this variable.
- Equivalised household income—The impact of household income depends on the size of the household. For this reason, income was equivalised to the number of household residents using the Australian Bureau of Statistics “modified OECD” equivalence scale (Australian Bureau of Statistics 2009b). In this scale, the first adult in a household had a

¹ Other age cut-offs were considered but did not significantly change descriptive outcomes.

weight of one point, each additional adult is allocated 0.5 points and each child under the age of 15 is allocated 0.3 points; the household income is then divided by this factor. This value was then adjusted using the consumer price index (CPI) to equate with 2009 Australian dollars. Note that the dataset does not include a measure of individual income.

- Employment status—Full-time, part-time or casual, secondary school studies, tertiary studies, unemployed or none of these (although not specified, for this age demographic this is likely made up of home duties/stay-at-home parents). People were able to nominate more than one of these, but for ease of interpretation multiple responses were coded as single responses with the following priority: full-time work, secondary study, tertiary study, part-time work, unemployed, none of these.
- Living independently with children—If someone was living outside of the parental home in a household with children under the age of 15, they were assumed to be living independently with their own children. This variable did not consider whether or not they were living with a spouse or partner.
- Living with parents—The travel survey identifies whether someone is the child of the oldest household member. This was used to indicate whether the young person lived with their parents. VISTA also identified whether someone was a grandchild of the oldest person; these people were also considered to be living in a parental household. The data did not identify people living with non-parental guardians.
- Survey year—Entered as dummy variables relative to 1994. If the other variables in the model can account for all of the change in licensing between 1994 and 2009, then the survey year should not be statistically significant. However if survey year is significant this suggests that the variables in the model do not fully account for licensing changes. It should be noted that licensing requirements changed significantly in 2007 with the requirement that learner drivers document 120 h of supervised driving before they can test for a driving license (Senserrick 2007).
- Gender and age

Results

Trend analyses

Figure 2 shows the licensing rates by age group obtained from the travel survey data between 1994 and 2009. Although they are from a different source, the results are broadly consistent with those from Fig. 1. Compared to the 1990s, younger age groups (other than 18-year-olds) are less likely to have a license and older age groups are more likely to hold a license. Interestingly, between 1994 and 1999 young adult licensing rates were slightly increasing. This is consistent with international research that found licensing rates among young adults tended to peak in the 1990s (Kuhnimhof et al. 2012). It is also worth noting that within an age cohort, licensing increased over time (e.g. those aged 20–23 in 1994 were 35–38 in 2009; their licensing rate rose from 87 % to over 94 % in this time).

Table 1 shows a sample of demographic characteristics of the survey sample. The younger age group (18–23) is more likely than the older age group (24–30) to be working part-time, studying, living with parents and living in a household with three or more cars. The older age group is more likely to have a driving license, work full-time, have their own children and live in a household with one or more cars per adult. The proportion of young

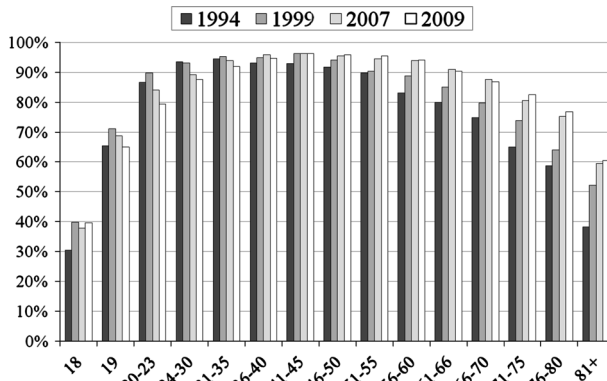


Fig. 2 Driver's licensing rates from survey data, 1994–2009

adults living with their parents was higher in the survey sample than the national average, where in 2006 23 % of people aged 20–34 lived with their parents (Australian Bureau of Statistics 2009a). This may be reflecting the nature of a household travel survey which is less likely to sample short-term leases, share-housing or dormitories where young adults may be living independently.

There are a few notable differences across the sample years:

As echoed in Fig. 2, driver's licensing dropped considerably between 1994 and 2009 for both age groups ($\chi^2(3) = 20.8$, $p < .001$ and $\chi^2(3) = 59.9$, $p < .001$).

Between the survey years, the 18–23 year-olds became more likely to work part-time, less likely to still be in secondary school and more likely to be in tertiary studies ($\chi^2(3) = 155.4$, $p < .001$, $\chi^2(3) = 23.0$, $p < .001$ and $\chi^2(3) = 17.3$, $p < .001$, respectively).

Between the survey years, the 24–30 age group became slightly more likely to be working part-time or undergoing tertiary studies and less likely to be outside the workforce/study ($\chi^2(3) = 33.3$, $p < .001$, $\chi^2(3) = 14.3$, $p < .01$ and $\chi^2(3) = 51.9$, $p < .001$, respectively).

Full-time employment remained fairly steady for both age groups and unemployment was slightly lower across the survey years ($\chi^2(3) = 49.4$, $p < .001$ and $\chi^2(3) = 27.7$, $p < .001$).

The younger age cohort became more likely to be living at home with parents ($\chi^2(3) = 36.6$, $p < .001$) and the older age cohort became less likely to live independently with their children ($\chi^2(3) = 40.4$, $p < .001$) across the survey years.

Average equivalised household income increased significantly across survey years despite being adjusted by CPI growth ($F(3,71455) = 4490.6$, $p < .011$), peaking in the 2007 sample.

Household car availability increased across the survey years ($\chi^2(3) = 85.4$, $p < .001$ and $\chi^2(3) = 18.1$, $p < .01$).

Interestingly, among the 24–30 group this sample did not follow the national trend of being more likely to live at home with parents. This may be due to the nature of the household travel survey method; it may be that young adults living independently are unlikely to respond to the travel survey.

Although this table gives some indication of aggregate demographic changes across survey years it cannot in itself explain changes in driver's licensing in that time. The next

Table 1 Survey sample demographics by survey year

	Age group	Survey year			
		1994	1999	2007	2009
Sample size	18–23	1529	923	2197	2102
	24–30	2015	1147	2442	2463
Holds a license (provisional or full)	18–23	75 %[2.2 %]	77 %[2.7 %]	74 %[1.8 %]	70 %[2.0 %]
	24–30	94 %[1.0 %]	93 %[1.5 %]	89 %[1.2 %]	88 %[1.3 %]
Employed full-time	18–23	32 %[2.3 %]	29 %[2.9 %]	31 %[1.9 %]	28 %[1.9 %]
	24–30	65 %[2.1 %]	65 %[2.8 %]	67 %[1.9 %]	65 %[1.9 %]
Employed part-time	18–23	9 %[1.4 %]	10 %[1.9 %]	12 %[1.4 %]	14 %[1.5 %]
	24–30	11 %[1.4 %]	11 %[1.8 %]	13 %[1.3 %]	14 %[1.4 %]
Secondary school studies	18–23	10 %[1.5 %]	11 %[2.0 %]	8 %[1.1 %]	7 %[1.1 %]
	24–30	0 %[0.0 %]	0 %[0.0 %]	0 %[0.0 %]	0 %[1.0 %]
Tertiary studies	18–23	39 %[2.4 %]	43 %[3.2 %]	44 %[2.1 %]	46 %[2.1 %]
	24–30	9 %[1.2 %]	13 %[1.9 %]	10 %[1.2 %]	11 %[1.2 %]
Unemployed	18–23	8 %[1.4 %]	5 %[1.4 %]	4 %[0.8 %]	5 %[0.9 %]
	24–30	6 %[1.0 %]	5 %[1.3 %]	4 %[0.8 %]	5 %[0.9 %]
Outside workforce/study ^a	18–23	3 %[0.9 %]	2 %[0.9 %]	2 %[0.6 %]	1 %[0.4 %]
	24–30	10 %[1.3 %]	7 %[1.5 %]	5 %[0.9 %]	6 %[0.9 %]
Lives away from parents and with own children	18–23	2 %[0.7 %]	1 %[0.6 %]	1 %[0.4 %]	1 %[0.4 %]
	24–30	21 %[1.8 %]	16 %[2.1 %]	14 %[1.4 %]	15 %[1.4 %]
Lives with parent(s)	18–23	71 %[2.3 %]	76 %[2.8 %]	79 %[1.7 %]	78 %[1.8 %]
	24–30	27 %[1.9 %]	30 %[2.7 %]	30 %[1.8 %]	27 %[1.8 %]
Average equivalised HH income (CPI adjusted, AU\$)	18–23	21,576 [818]	35,990 [1176]	73,031 [2190]	48,909 [1125]
	24–30	28,794 [908]	46,175 [1443]	87,491 [2634]	61,846 [3455]
No cars	18–23	6 %[1.0 %]	6 %[1.5 %]	4 %[0.8 %]	4 %[0.8 %]
Less than one car per adult		69 %[2.3 %]	62 %[3.1 %]	58 %[2.1 %]	59 %[2.1 %]
One or more cars per adult		25 %[2.2 %]	32 %[3.0 %]	38 %[2.0 %]	38 %[2.1 %]
No cars	24–30	5 %[1.0 %]	6 %[1.4 %]	5 %[0.9 %]	6 %[0.9 %]
Less than one car per adult		43 %[2.2 %]	38 %[2.8 %]	38 %[1.9 %]	39 %[1.9 %]
One or more cars per adult		53 %[2.2 %]	56 %[2.9 %]	57 %[2.0 %]	56 %[2.0 %]

95 % Confidence interval shown in brackets

^a Although not specified in all travel survey years, this designation likely contains a high proportion of home duties/stay-at-home parents

section presents a set of binary logistic regression models to examine the disaggregated associations between these variables and licensing.

Binary logistic regression models

Two binary logistic regression models were run, one including household vehicles and one without it.

Table 2 presents the results of these models. Both models were statistically significant at 95 % confidence. The model including household vehicles was slightly more powerful

Table 2 Logistic regressions predicting young adult licensing, with and without household vehicles

	With household vehicles Model A		Without household vehicles Model B	
Sample N	14,517		14,538	
Chi-square omnibus test	3975.8		2652.0	
Cox and Snell pseudo-R ²	0.240		0.167	
Nagelkerke pseudo-R ²	0.397		0.276	
−2 Log likelihood	9472.9		10,790.5	
	Regression weight	Odds ratio	Regression weight	Odds ratio
Constant	−3.32		−1.80	
Ratio of cars to adults				
None	Reference	1	n/a	n/a
Less than one	1.49*	4.42 [3.67–5.34]	n/a	n/a
One or more	3.43*	30.76 [24.60–38.47]	n/a	n/a
Household income quartile				
Lowest	Reference	1	Reference	1
Second	0.12	1.13 [0.97–1.31]	0.37*	1.45 [1.26–1.66]
Third	0.28*	1.33 [1.13–1.57]	0.70*	2.02 [1.74–2.35]
Fourth	0.51*	1.66 [1.39–1.99]	1.01*	2.74 [2.32–3.24]
Employed full-time	Reference	1	Reference	1
Employed part-time	−0.69*	0.50 [0.42–0.60]	−0.79*	0.45 [0.38–0.53]
Secondary school	−2.65*	0.07 [0.06–0.09]	−2.81*	0.06 [0.05–0.08]
Tertiary studies	−0.60*	0.55 [0.48–0.63]	−0.87*	0.42 [0.37–0.48]
Unemployed	−1.34*	0.26 [0.21–0.32]	−1.59*	0.20 [0.17–0.25]
Not in workforce	−1.91*	0.15 [0.11–0.19]	−1.94*	0.14 [0.11–0.19]
Has own children	0.59*	1.89 [1.38–2.34]	1.05*	2.87 [2.25–3.67]
Lives with parent(s)	0.47*	1.59 [1.40–1.81]	0.65*	1.92 [1.70–2.16]
Female	−0.20*	0.82 [0.74–0.91]	−0.14*	0.87 [0.79–0.96]
Age	0.16*	1.18 [1.15–1.20]	0.17*	1.18 [1.16–1.20]
Year				
1994	Reference	1	Reference	1
1999	−0.04	0.96 [0.79–1.17]	−0.16	0.86 [0.71–1.03]
2007	−0.75*	0.47 [0.40–0.56]	−0.81*	0.45 [0.38–0.52]
2009	−0.93*	0.40 [0.34–0.46]	−0.94*	0.39 [0.34–0.45]

Brackets after odds ratios indicate 95 % confidence interval

* Values marked with are statistically significant at 95 % confidence

with pseudo-R² values of 0.240 and 0.397²; for the model without household vehicles these values were 0.167 and 0.276. The strength of these R² values is reasonable, although there are still important influences not captured by the model.

In the model that includes household car ownership (Model A), these values exhibited a strong relationship to license-holding. This effect has been found in previous literature and

² Pseudo R² statistics are applied in logistic regression, logit and probit models and are interpreted in a similar fashion to the R² values reported with linear regression models.

at a similar effect size (Licaj et al. 2012). In both models higher household income increased the likelihood of a young adult having a driving license, although this effect was stronger in Model B. As demonstrated in previous research, clearly income and car ownership are interrelated variables.

The next most influential variables in both models were the employment and study variables, with similar effects present in both models. Compared to being employed full-time, young adults in all other employment or study situations were less likely to have a license. Being in secondary school had the strongest negative association with licensing, followed by (in order of magnitude) being outside the workforce, being unemployed, tertiary studies and being employed part-time. These associations were of similar magnitude across both models.

In both models, living independently with children significantly increased the likelihood that a young adult would have a license, although the effect was much larger in Model B. Similarly in both models living with parents increased the likelihood that a young adult would have a license.

Even when controlling for employment and other demographics, women were slightly less likely to have a license than men and each additional year of age increased the likelihood of licensing. Note that in contrast with recent work that found that young men are more likely to be reducing their car use than young women (Kuhnimhof et al. 2012), in this dataset both men's and women's rates of licensing have fallen.

Finally, the influence of survey year was statistically significant. The difference between 1994 and 1999 was not significant, which means that any difference in licensing between these years could possibly be explained by changes in the variables in the model. However the latter two survey years were statistically significant in both models. This means that even when controlling for all other variables in the model, young adults were still less likely to hold a license in 2007 or 2009 than in 1994. The possible reasons for this are discussed in the conclusions.

Discussion and conclusions

This paper presents some of the first research empirically exploring links between declining licensing rates of young adults and trends in employment, education and living arrangements. Table 3 summarises the effects found in this study in the context of nationwide demographic trends.

Even when controlling for household income, there was a significant relationship between the employment status of a young adult and whether or not they had a license; this is likely to be because part-time employment and studying result in less personal disposable income. In recent decades, young adults have become less likely to work full-time and more likely to work part-time or study; taken together, these findings suggest that this shift may have a significant impact on the trend toward reduced young adult licensing.

Although employment and study are linked to broader economic trends, it is unlikely that wider economic factors are having macro-level impact on licensing in this dataset. In Australia, economic conditions significantly improved and youth unemployment declined between 1994 and 2007, with only a minor downturn in 2008 (Australian Bureau of Statistics 2013). This supports research from other countries that finds youth licensing trends tend to decline *before* recessions begin (Delbosc and Currie 2013b). Macro-economic trends may play a role in licensing changes, but the impact has yet to be fully explored.

Table 3 Declining licensing in young adults—potential causes from living arrangement trends and measured licensing impacts

Demographic factor	Trend among young adults ^a	Modelled association with licensing	Theorised combined impact
Full-time employment	Decrease	Increases likelihood	Less disposable income for young adults may cause a net decline in driver licensing
Part-time employment	Increase	Decreases likelihood	
Educational participation	Increase	Decreases likelihood	
Cohabitation, marriage, child-rearing	Decrease	Increases likelihood	Postponement of living independently with children reduces pressure on young people to seek driver license
Living with parents	Increase	Possibly increases likelihood	Living with parents provides greater access to vehicles but may also be a proxy for limited economic independence

^a These trends are occurring nationally in Australia and in many other countries, as documented in the Research Context section and (Delbosc and Currie 2013b)

There was a significant relationship between increased licensing rates and living independently with one's own children. Nationally there are small but significant trends toward young adults postponing cohabitation, marriage and having children until later in life. These findings suggest that this may be another influence on the trend toward reduced licensing among young adults.

Another national demographic trend is the increase in young adults living with their parents. In this study living at home with parents was associated with *higher* licensing rates, perhaps because living the parental home proves opportunities for supervised driving and shared household vehicles. However it is important to note that in exploratory modelling of this data (Delbosc and Currie 2013a), this coefficient was sometimes negative, suggesting that living with parents *decreased* licensing. The instability of this effect is a strong note of caution; clearly this effect requires further research before firm conclusions can be reached.

Taken together, these results suggest that having a driving license is associated with an 'adult' life stage represented by full-time employment, having children and having higher incomes. Nationally and internationally, a growing number of young adults are delaying some or all of these transitions, which in turn reduces the need for (and the means to pay for) a driving license for these young adults.

The implications of this finding are important for urban planners. The trend toward higher education, part-time employment, delaying marriage and delaying child-bearing shows no sign of reversing. As long as these trends continue, it is likely that an increasing number of young adults will delay or forgo getting a license. Urban planners should be ready to support this trend by providing alternative transport options and providing housing stock attractive to young people that is accessible to jobs and universities. Although these young adults may one day start families, take-up a license and move to the suburbs, each year they are able to meet their travel needs without a license means fewer cars on the road.

These life stage decisions—such as car licensing and ownership, living arrangements, working arrangements and family planning—are complex and likely to be considered somewhat simultaneously. This presents complexities that a simple regression model may

not fully capture, suggesting that future research should consider alternative modelling strategies such as structural equation modelling.

The regression model also showed that there were still important factors that contribute to licensing that could not be captured by the existing variables: the pseudo- R^2 values were relatively modest and the variables representing the later survey years were statistically significant. This indicates that there are other factors influencing licensing not captured in the present model. A likely explanation is that in 2007 graduated driver licensing was introduced in Victoria; most notably learner drivers are now required to document 120 h of supervised driving (Senserrick 2007).³

It is unfortunate that due to data limitations the model could not include variables measuring urban density or transit access as past research has suggested that this can influence licensing rates (McDonald and Trowbridge 2009). The cost of motoring is also a likely factor although its impact is difficult to measure; although gasoline prices have increased in real terms in recent years, the overall cost of motoring has declined in many countries (Noble 2005; Davis et al. 2012). Similarly, broader economic factors have been cited as a potential explanation although licensing began its decline years before the global financial crisis of 2008 (Berg 2001; Davis et al. 2012). Furthermore, there is an emerging body of research suggesting that the attitudes of the younger generation toward cars are fundamentally shifting. It may be that for young people, cars are no longer the status symbol that they once were, with technology replacing cars as the new object of desire (Kalmbach et al. 2011). Further research is needed to better understand these diverse influences on licensing decisions and the implications for transport planning into the future.

Acknowledgments The authors would like to thank the Victorian Department of Transport for making the VATS and VISTA data available for analysis. An earlier form of this paper was presented at the 92nd Transportation Research Board Annual Meeting (Delbosc and Currie 2013a). Any errors or omissions are the responsibility of the authors.

References

- Australian Bureau of Statistics: Home and away: the living arrangements of young people. Australian Social Trends. Report 4102.0 (2009a)
- Australian Bureau of Statistics: Household income and income distribution, 2007–2008 (2009b)
- Australian Bureau of Statistics: Population by age and sex, Victoria. 3101.0 (2010)
- Australian Bureau of Statistics: Education and work, Australia (1991–2011). Report 6227.0 (2011)
- Australian Bureau of Statistics: Australian social trends 2012: locve me do, commonwealth of Australia. Report 4102.0 (2012)
- Australian Bureau of Statistics: Labour force, Australia. Report 6202.0 (2013)
- Berg, H.-Y.: Understanding Subgroups of Novice Drivers: A Basis for Increased Safety and Health. Linköping University, Linköping (2001)
- Bureau of Labor Statistics: “School’s out”. Spotlight on statistics Retrieved 10 October 2012, from http://www.bls.gov/spotlight/2011/schools_out/ (2011)
- Cobb-Clark, D.A.: Leaving home: what economics has to say about the living arrangements of young Australians. Aust. Econ. Rev. **41**(2), 160–176 (2008)
- Davis, B., Dutzik, T., Baxandall, P.: Transportation and the New Generation: Why Young People Are Driving Less and What It Means for Transportation Policy. Frontier Group, Santa Barbara (2012)
- Delbosc, A., Currie, G.: Using online discussion forums to study attitudes toward cars and transit among young people in Victoria. 35th Australasian Transport Research Forum, Perth (2012)

³ To test whether this factor has a further effect on other variables, models were run to test for an interaction effect with survey year. There were no significant interaction effects.

- Delbosch, A., Currie, G.: Are changed living arrangements influencing youth driver license decline? Transportation Research Board 92nd Annual Meeting, Washington, DC (2013a)
- Delbosch, A., Currie, G.: Causes of youth licensing decline: a synthesis of evidence. *Transp. Rev.* **33**(3), 271–290 (2013b)
- Federal Highway Administration: Licensed drivers, by sex and age group, 1963–1995 (Table DL-220) (2011)
- Gray, M., Qu, L., Weston, R.: Fertility and Family Policy in Australia. Australian Institute of Family Studies, Melbourne (2008)
- Kalmbach, R., Bernhart, W., Kleimann, P.G., Hoffman, M.: Automotive Landscape 2025: Opportunities and Challenges Ahead. Roland Berger, Munich (2011)
- Kuhnimhof, T., Armoogum, J., Buehler, R., Dargay, J., Denstadli, J.M., Yamamoto, T.: Men shape a downward trend in car use among young adults—evidence from six industrialized countries. *Transp. Rev.* **32**(6), 761–779 (2012)
- Kuhnimhof, T., Buehler, R., Dargay, J.: A new generation: travel trends for young Germans and Britons. *Transp. Res. Rec.* **2230**, 58–67 (2011)
- Le Vine, S., Jones, P.: On the Move: Making Sense of Car and Train Travel Trends in Britain. RAC Foundation, London (2012)
- Le Vine, S.E., Jones, P.M., Polak, J.W.: Has the historical growth in car use come to an end in great britain? 37th European Transport Conference. Leeuwenhorst, Netherlands (2009)
- Licaj, I., Haddak, M., Pochet, P., Chiron, M.: Individual and contextual socioeconomic disadvantages and car driving between 16 and 24 years of age: a multilevel study in the Rhône Département (France). *J. Transp. Geogr.* **22**, 19–27 (2012)
- McDonald, N., Trowbridge, M.: Does the built environment affect when American teens become drivers? Evidence from the 2001 National Household Travel Survey. *J. Saf. Res.* **40**, 177–183 (2009)
- Newman, P., Kenworthy, J.: ‘Peak car use’: understanding the demise of automobile dependence. *World Transp. Policy Pract.* **17**(2), 31–42 (2011)
- Noble, B.: Why are some young people choosing not to drive? European Transport Conference, Strasbourg (2005)
- Office for National Statistics: Young people in the labour market—2011. London (2011)
- Preusser, D.F., Ferguson, S.A., Williams, A.F., Leaf, W.A., Farmer, C.M.: Teenager driver licensure rates in four states. *J. Saf. Res.* **29**(2), 97–105 (1998)
- Prevedourous, P.D., Schofer, J.L.: Factors affecting automobile ownership and use. *Transp. Res. Rec.* **1364**, 152–160 (1992)
- Puentes, R., Tomer, A.: The Road... Less Traveled: An Analysis of Vehicle Miles Traveled Trends in the U.S. Metropolitan Infrastructure Initiatives Series. Brookings Institution, Washington DC (2008)
- Raimond, T., Milthorpe, F.: Why are young people driving less? trends in license-holding and travel behaviour. Australasian Transport Research Forum, Canberra (2010)
- Ryan, J.M., Han, G.: Vehicle-ownership model using family structure and accessibility application to Honolulu, Hawaii. *Transp. Res. Rec.* **1676**, 1–10 (1999)
- Scheiner, J., Holz-Rau, C.: Travel mode choice: affected by objective or subjective determinants? *Transportation* **34**(4), 487–511 (2007)
- Senserrick, T.M.: Recent developments in young driver education, training and licensing in Australia. *J. Saf. Res.* **38**(2), 237–244 (2007)
- Sivak, M., Schoettle, B.: Recent changes in the age composition of U.S. drivers: implications for the extent, safety, and environmental consequences of personal transportation. *Traffic Inj. Prev.* **12**, 588–592 (2011)
- Sivak, M., Schoettle, B.: Recent changes in the age composition of drivers in 15 countries. *Traffic Inj. Prev.* **13**, 126–132 (2012)
- Taylor, P., Parker, K., Kochhar, R., Fry, R., Funk, C., Patten, E., Motel, S.: Young, Underemployed and Optimistic: Coming of Age, Slowly, in a Tough Economy. Pew Research Center, Washington, D.C (2012)
- United States Census: National estimates by age, sex, race: 1900–1979 (2004)
- VicRoads: VicRoads license breakdown by age, 2001–2011 (2012)
- Woldeamanuel, M.G., Cyganski, R., Schulz, A., Justen, A.: Variation of households’ car ownership across time: application of a panel data model. *Transportation* **36**, 371–387 (2009)
- Zumkeller, D., Chlond, B., Manz, W.: Infrastructure development in Germany under stagnating demand conditions: a new paradigm? *Transp. Res. Rec.* **1864**, 121–128 (2004)

Author Biographies



Alexa Delbosc is a research fellow in the Public Transport Research Group of the Institute of Transport Studies. Her research focuses on the changing travel habits of young people, transport psychology, human factors in public transport and the use of powered-two-wheel transport.



Graham Currie is a Professor of Public Transport and Director of the Public Transport Research Group at the Institute of Transport Studies (ITS), Monash University. He directs the World Transit Research Clearinghouse (www.worldtransitresearch.info) and has published over 200 research papers in leading research journals and conferences.

Copyright of Transportation is the property of Springer Science & Business Media B.V. and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.